

Priya needs an answer by 3pm.

Two weeks in, and the working session from Level 1 actually happened — Maya asked Growth, Product, and Finance to agree on one definition of "active." They did. But this morning, Priya's dashboard says signups are up **40%** this month, and Farrah's spreadsheet says they're flat. You have real database access now. Time to find out who's right yourself — not by asking whose Slack message sounds more confident.

UNLOCKS: READ ACCESS TO SOLACE'S DATABASE

1

2

3

4

5

6

7

8

Database access granted

IT set you up with read-only access this morning. Three tables you'll actually use this level:



signups

One row per person who started a signup flow. Includes a **status** column: trialing, active, paused, cancelled.



subscriptions

One row per billing period. This is Farrah's source of truth for revenue.



sessions

One row per app open. This is what Growth's original "active" definition from Level 1 was built on.

Keep this page open. Every query this level touches one of these three tables.

Wrong filter, wrong answer

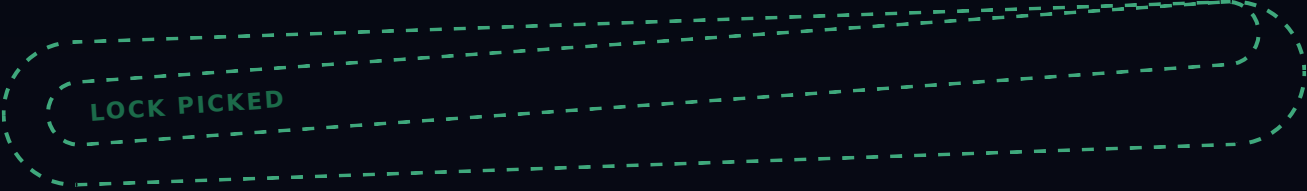
LOCKED — you can't move to the next mission until you find the bug.

Here's the exact query behind Priya's dashboard:

```
SELECT COUNT(*) AS signups
FROM signups
WHERE status != 'cancelled'
AND signup_date >= '2026-06-01';
```

What's wrong with this filter? Write it before you turn the page.

No peeking — the bug's explained on the next page, not this one.



LOCK PICKED

Why the filter let everyone in

"!= 'cancelled'" catches everyone who *isn't* cancelled — including trialing and paused signups that haven't converted yet. Priya's 40% jump is mostly trial signups, not paying customers. A negative filter (!=) lets in anything you didn't think to exclude. A positive filter (= 'active') only lets in exactly what you meant.

Your first real query

Fix Priya's query yourself. Fill in the three blanks below.

```
SELECT COUNT(-----) AS signups
FROM signups
WHERE status = -----
      AND signup_date >= -----;
```

Blank 1 — how do you make sure the same person isn't counted twice?

Blank 2 — which exact status means "converted," not just "not cancelled"?

Blank 3 — the date this month's window starts.



Try it live. Scan to open the SQL Playground on the site and run real queries against a live database — same engine, your own pace.

Marketing says up 40%. Finance disagrees.

CONFIDENTIAL

Priya's dashboard: **42,300** signups this month, up from **30,200** last month. Farrah's spreadsheet: **31,500** signups this month — roughly flat. Same company, same month, two very different stories.

Both numbers came from a query. One of them just doesn't know it yet.

Before you look at Farrah's query

Farrah's spreadsheet number came from a query too — she just doesn't call it that. Here's her formula, translated to SQL:

```
SELECT COUNT(DISTINCT user_id)
FROM signups
WHERE status = 'active'
AND signup_date >= '2026-06-01';
```

Predict: whose query is closer to "the truth" — Priya's or Farrah's? Why?

Lock in your guess — the answer's waiting on the next page.

VERDICT IN

Farrah's query wins — here's why

Farrah's is closer — she's filtering on the exact state she means (active), and deduping with DISTINCT. Priya's dashboard was built months ago, before Solace added the "paused" status, and nobody updated the filter. This isn't Priya's fault — it's what happens when a query doesn't get revisited as the business changes.

REFLECTION CARDS

Before you move on

Where else in your own work might a filter be quietly stale, like Priya's dashboard?

Why does DISTINCT matter here — what would inflate the count without it?

Is Farrah "right" because her number is smaller, or for a different reason?

From someone who's seen this before

Dashboards rot. Nobody budgets time to revisit a query after it ships, and a "temporary" filter becomes permanent silently.

The first thing I do when two numbers disagree isn't pick a side — it's open both queries side by side. Usually one of them is just old.

Meanwhile, in the #eng-random Slack channel:

R

Ravi

lol priya's dashboard again

S

Sam

yeah classic stale filter. anyway are you doing the panel thing thursday

R

Ravi

which panel

S

Sam

the one where they always ask "walk me through a metric you found was wrong, and how you caught it" — swear every interviewer asks some version of that

R

Ravi

oh yeah. ok lunch?

S

Sam

lunch

BOSS FIGHT

Present Your Findings to Growth

Priya's whole team is in the room, expecting you to just confirm her 40% number. Instead, you have to tell them it's wrong — in front of the person whose dashboard it is.

☐ **A — Tell them the number's wrong and move on.**

Trade-off: fast, but doesn't help Priya trust her own dashboards going forward.

☐ **B — Show them exactly which filter broke and why, then offer to fix the dashboard query together.**

Trade-off: takes longer in the room, but Priya leaves knowing how to catch this herself next time.

☐ **C — Quietly fix the dashboard yourself overnight and say nothing.**

Trade-off: fastest for you, but Priya never learns why it broke — it'll happen again with a different filter.

B is the one that actually builds Priya's trust in her own tools, not just in you. Same lesson as the Level 1 Boss Fight: the fix that teaches beats the fix that just resolves the meeting.

SQL Cheat Cards — Set 1

SELECT & WHERE Basics

SELECT the columns you need, not SELECT *. WHERE filters rows before any grouping happens.

Positive vs. Negative Filters

Prefer = / IN over != / NOT IN when you know the exact state you want. Negative filters silently let in anything you forgot to exclude.

COUNT(*) vs COUNT(DISTINCT col)

COUNT(*) counts rows. COUNT(DISTINCT user_id) counts people. They stop being the same the moment a table can have more than one row per person.

Date Filters

>= and < are safer than BETWEEN for date ranges. Watch for timezone boundaries and off-by-one month errors.

Business-card sized on purpose — print these, keep them on your desk.

Level 2, condensed

The bug: "!= 'cancelled'" included trialing and paused signups Priya never meant to count.

The fix: filter on the exact state (= 'active') and dedupe with COUNT(DISTINCT user_id).

The real numbers: 42,300 (Priya, unfiltered) vs. 31,500 (Farrah, correct) — a 40% jump that was mostly noise.

The Boss Fight call: show the bug and fix it together, don't just quietly patch it.

Test Yourself on Solace's Data

Three questions, one real answer each. No skipping ahead, no peeking at the next page early. Pick your answers below, then turn the page to see how you did.

1. Which filter accidentally included trialing and paused users?

- ☐ A. status = 'active'
- ☐ B. status != 'cancelled'
- ☐ C. signup_date >= '2026-06-01'

2. What does COUNT(DISTINCT user_id) protect against?

- ☐ A. Wrong date range
- ☐ B. Counting the same person twice
- ☐ C. Typos in status values

3. Why was Priya's dashboard wrong, even though nobody had touched it recently?

- ☐ A. She wrote it wrong originally
- ☐ B. The business changed and the filter didn't update
- ☐ C. The database was down

Your verdict — and the answer key — is one page away.

CASE FILE VERIFIED

How'd you do?

1 → B. "!= 'cancelled'" let in trialing and paused signups that were never meant to count.

2 → B. Without DISTINCT, a person with more than one signup row gets counted more than once.

3 → B. Solace added a "paused" status after the dashboard was built, and the filter never caught up.

Got all 3?

You're already thinking like the person who catches the stale filter before it hits a Growth all-hands. That instinct is the whole job.

Got 2 out of 3?

Solid — you're reading the query, not just the dashboard number. Flip back to Page 4 and Page 8 for the one that slipped past.

Got 1 or fewer?

No penalty for that in Level 2: The SQL Forest — the whole level exists to build this exact instinct. Re-read the Cheatsheet on Page 13 and carry it into Level 3.

The Signups Reconciliation

No answer key here — reconcile a real discrepancy from your own life or work, the way you just reconciled Priya's and Farrah's numbers.

Your Case

Two conflicting numbers	Where each came from	Which filter/ definition differs	Which one you'd trust, and why
CLUE #1 — What are the two numbers, and who owns each one?			
CLUE #2 — Which filter or definition is the real difference?			
CLUE #3 — Which one would you actually trust in a room, and why?			

This one goes on your Career Checklist too — a second real audit artifact for your Level 8 portfolio project.

LEVEL COMPLETE



World 2: The SQL Forest

You finished Level 2 and wrote your first production-shaped query.

Priya asked for your Slack handle directly instead of going through Maya. Farrah still isn't fully convinced — but she stopped saying "I don't trust any of them."

"A dashboard is only as current as the last time someone remembered to check it."

NEXT UP

Level 3: The JOIN Labyrinth

You've earned real database access. Next, you find out what happens when two of Solace's tables tell different stories about the same customer.